AMAZON STORAGE S3

1. Simple Storage Service (s3) object level storage Access from anywhere
2. Elastic File System (EFS) mostly used for the Linux based system
3. Elastic Block Level Storage (EBS)
4. Glacier new name s3 glacier
5. Snowball used for the data migration

➢ Block Level Storage

Block storage is suitable for transaction database random read/write loads and structured database storage

Block storage divides the data to be stored in evenly sized block data chunk for instance a file can be split into evenly sized blocks before it is stored

Data block Stored in Block storage would not contains metadata (data about data created data modified content types data)

Block storage only keeps the address (index) where the data blocks are stored it does not care what is in that block just how to retrieve it when required

➢ Object level Storage

Object Storage Storage Stores the files as a whole and does not divide them

In object storage an object is

- The file / data itself
- its metadata
- object global unique ID

The Object Global Unique ID is a Unique Identifier for the Object (can be the object name itself)
And it must be Unique such that it can be Restrived disregarding where its physical

Object storage can't be mounted as a drive
Example of object storage solutions Dropbox AWS S3 facebook

➢ Simple Storage Service   S3

S3 is a storage for the internet it has a simple web Services interface for simple storing & Retrieving of any amount of data anytime from anywhere on the internet

S3 is object Based Storage
You can't install operating system on s3
S3 has a distributed data – store architecture where object are redundantly stored in multiple location (min 3 location is same region)

Data is Stored in Bucket
A Buckets is flat Container of Object
Max Capacity of a Buckets is 5 TB

You can create folders in your Buckets (Available through console)

You can't create Nested Buckets
Buckets Ownership is Non-transferrable
S3 Buckets is Region Specific

You can have up to 100Buckets per accounts (may expand on Request)

- **S3 buckets Naming Rules**

S3 Buckets names(keys) are Globally Unique across all AWS Regions

Buckets Name Can’t be change after they are created

If a buckets is deleted its name becomes available again to you or other account to use

Buckets name must be at least 3 and no more then 63 characters

Buckets name are part of the url use to access bucket

Buckets name must be a Series of one or more labels (xyz.bucket)

Buckets name can contains lowercase number and hyphen can’t use uppercase letter

Bucket name should not be an IP

Each label must start and end with a lowercase or a number

By default buckets and its object are private by default only owner can access the bucket

S3 buckets – Sub resources

Sub resources for S3 buckets include :-

Lifecycle :- To decide on Object Lifecycle management

Website :- To hold Configuration related to static website hosted in S3 Buckets

Versioning :- Keep Object versions as it changes (gets Updated)
once we enable the versioning we can’t disables versioning

Access control list :- Bucket Policies

The name is simply two parts :-

Bucket Regions endpoint / Bucket name

Example ://s2-eu-web1.amazonaws.com/mybucket

S3 object

An object size stored in an S3 Buckets can be 0 byte to 5 TB

Each object is stored and Retrieved by an Unique key (Id or name)

An Object in Aws S3 is Uniquely identified and addressed through service endpoint

- Service endpoint
- Buckets key (name)
- Object key (name)
- Optionally Object Version

Object stroed in a S3 buckets in a Region will never leave that Region unless you Specifically move them to another Region CRR

A buckets Owner can grant cross – account permission to another AWS account (or urser in another accounts) to upload objects

You can grant S3 buckets / Object Permission to ::

- Individual users
- AWS Account
- Make the Resource Public
- Or to all authenticate users

➢ S3 Bucket Versioning

Buckets Versioning is a S3 Buckets Sub – Resource used to protected against accidental Object/data deletion or Overwrites

Versioning can also be used for data Retention can Archive

Once you enable Versioning on a Buckets it can't be disabled however it can be suspended

When enabled Bucket Versioning will protect existing and new object and maintains their version as they are updated

Updating copy delete action an objects

When versioning is enabled and you try to delete an object a delete marker is placed on the object

you can still view the object and the delete marker

If you Reconsider deleting the Object you can delete the “Delete Marker “ and the Object will be Available again

You will be charged for all s3 storage cost for all object versions stored

You can versioning with s3 lifecycle policies to delete older version or you can move them to a cheaper s3 storage (or glacier)

Bucket Versioning State :-

- Enable
- suspende

Versioning applies to all object in a bucket and not partially applied

Object existing before anabling versioning will have a version id or “null “

If you have a buckets that is already versioned then you suspend versioning existing object and their versions remain as it is …

However they will not be updated/versioned further with future updates
versioned further will future updated while the buckets versioning is suspended

New object (uploaded after suspensions) they will have a version id “Null “

If the same key (name) is used to store another object it will override the existing one

An object Deleting in a suspended versioning buckets will only delete the object with id “null “

- **S3 Bucket Versioning – MFA delete**

Multifactor authentication delete is a versioning capacity that adds another levels of security in case your account is compromised

This add another layers of security for the following

Permanent deleting an object

MFA delete Requires :-

You Security Credentials

This Code Displayed on an Approved Physical or s/w Based Authentication device

- **S3 Multipart Upload**

Is used to upload an object In parts

Parts are uploaded independently and in parallel in any order

It IS Recommended for object sized of 100mb or large (min size 5 mb)

You must used for object larger than 5gb

This is done through s3 multipart upload Api

- **Copying s3 object**

The Copy Operation Creates a copy of an Object that is Already Stored in Amazon S3

You can create a copy of your object up to 5gb in size a single atomic operation

How ever to copy an object greater than 5gb you must use the multipart upload Api

Incur charges if copy to another region

Use the copy operation to

Generate additional copies of the subject

Renaming object (copy to new Name)

Changing the copy’s storage class or encrypt it at rest

Move object across AWS location/Region

Class Object Metadata

➢ CLASSES OF AMAZON S3

1 . Amazon s3 standard
2 . Amazon s3 glacier deep archive
3 . Amazon glacier
4 . Amazon s3 standard infrequent access
5 . Amazon s3 one Zone IA
6 . Amazon s3 intelligent Tiering

➢ Amazon S3 Standard

S3 standard offers high durability availability and performance object storage for frequently accessed data

Durability is 99.9999999%
Designed for 99.99 availability over a given year

Support SSL for data in transit and encryption of data at rest
the storage cost for the object us fairly high but there is very less charge for accessing the object
Largest object that can be uploaded in single put is 5gb

Amazon S3 IA

S3 – IA is for data that is accessed less frequently but requires rapid access when needed

The storage cost is much cheaper than s3 S3- standard almost half the price but you are charged more heavily for accessing your data (object)

Durability is 99.999999%
Resilient against events that impact an entire az

Availability is 99.9 % in year

Support SSL for data in transit and encryption of data at rest

Data that is deleted from s3-IA charged for a full 30days

Backed with the amazon s3 service level agreement for availability

Amazon S3 Intelligent tiering

S3 – Intelligent tiering storage class is designed to optimized cost by automatically moving data to the most cost effective access tier

It work by storing object in two access tiers

If an object in the infrequent access tier is accessed it is automatically moved back to the frequent access tier

There are no retrieval fees when using the s3 intelligent tiering storage class and no additional tiering fee when object are moved between access tiers

Same low latency and high performance of s3 standard object less than 128kb cna't moved to IA

Durability is 99.999999%

Availability is 99.9 % in year

➢ Amazon one – zone IA

S3 – ONE Zone – IA is for each that is accessed less frequently but Requires rapid access when needed data stored in single az

Ideal for those who want lower cost option of iA – data

IT is good choice for storing secondary backup copies of on premise data or easily re creatable data

You can use s3 lifecycle policies
Durability is 99.999999 %
Availability is 99.99999 %

Because s3 one zone – IA stores data in single AZ data stored in this storage class will be lost in the event of Az destruction

Amazon S3 Glacier

S3 – Glacier is a secure durable low cost storage class for data archiving

To keep cost low yet suitable provide three Retrieved options that range from a few minutes to hours

You can upload object directly to glacier or use lifecycle policies

Durability is 99.99999 %
Data Resilient in the event of one entire AZ destruction

Support SSL for data in transit and encryption data at rest

You can Retrieve 10gb of your amazon s3 glacier data per month for free with free tier account

- **Amazon S3 Glacier Deep Archive**

S3 – Glacier Deep Archive is Amazon s3 cheapest Storage

Design is Retain data for long period Eg – 10 year

ALL Object stored in S3 glacier Deep archive are Replicated and stored across at least at three

GEOGRAPHICALLY dispersed AZ

Durability is 99.999999%
Ideal Alternative to magnetic tape libraries

Storage cost is up to 75% less than for the existing s3-glacier storage class

Availability is 99.99 %